Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

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June 11th

⌛ (Badly and Slowly) Approximating the Score Function using Optimal Transport

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The main approaches are based on the continuity equation

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0 \tag{1}$$

GOAL: Given μ (unknown) and γ (easy to sample from), **build/find** v_t such that the corresponding ρ_t evolve from μ to γ according to (1).

Score-based diffusion models

Let $v_t = -\nabla(\ln \rho_t + V)$ be fixed. The **score** at time t is $\nabla \ln \rho_t$. The continuity equation becomes

$$\partial_t \rho_t = \Delta \rho_t + \operatorname{div}(\rho_t \nabla V). \tag{2}$$

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- train a neural network that tries to directly approximate the score;
- or train a neural network $x_{0,\theta}$ that tries to denoise the data, then the score is given by the Tweedie formula:

$$\nabla \ln \rho_t(x_t) \approx \frac{\alpha_t x_{0,\theta}(x_t, t) - x_t}{\sigma_t^2}.$$

Using OT to learn the score

Consider $v_t = -\nabla(\ln \rho_t + V)$ where $V(x) = \frac{\|x\|^2}{2}$. Then, the Fokker-Planck equation

$$\partial_t \rho_t = \Delta \rho_t + \mathrm{div}(\rho_t \nabla V).$$

[1] R. Jordan, D. Kinderlehrer, and F. Otto. "The Variational Formulation of the Fokker–Planck Equation". en. In: *SIAM Journal on Mathematical Analysis* 29.1 (Jan. 1998), pp. 1–17. ISSN: 0036-1410, 1095-7154. DOI: 10.1137/S0036141096303359. URL: http://epubs.siam.org/doi/10.1137/S0036141096303359.

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- transports any initial condition $\rho_0 \in \mathcal{P}(\mathbb{R})$ towards the standard Gaussian $\gamma = \mathcal{N}(0, Id)$ and
- can be seen as the Wasserstein gradient flow of the energy[1]

$$\mathcal{F}(\rho) := \int V\rho + \int \rho \ln \rho$$

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Using OT to learn the score

We can discretize in time the gradient flow using implicit Euler scheme (or **JKO scheme**):

$$\rho^\tau_{n+1} := \underset{\rho \in \mathcal{P}_2(\mathbb{R}^d)}{\arg\min} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2(\rho, \rho^\tau_n)^2 \right\} \tag{3}$$

[2] F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer, Oct. 2015. ISBN: 978-3-319-20827-5.

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As explained by Santambrogio in his book[2], we have

$$\nabla \psi_n^{c_\tau} = v_{n\tau} = -\nabla(\ln \rho_{n+1}^{\tau} + V),$$

where $\psi_n^{c_\tau}$ is a Kantorovich potential related to the transport from ρ_n^{τ} to ρ_{n+1}^{τ}. Therefore approximating the score amounts to approximating $\nabla \psi_n^{c_\tau}$.

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Stochastic Gradient Descent

Theorem 1

If $\alpha_k = \frac{R}{B\sqrt{k}}$, then

$$\forall T \geq 1, \quad \mathbb{E}\left[G_n^\tau(\bar{\theta}_T^n) - G_n^\tau(\theta_n^\tau)\right] \leq BR\left(\frac{2}{T} + \frac{1}{\sqrt{T}}\right),$$

where $\bar{\theta}_T^n$ denotes the arithmetic mean of the family $(\theta^{(k)})_{k=1}^T$ and B depends only on R and $\sup_x \|\phi(x)\|$.

Some animations

Click here to play video

Thank you for your attention!